

Iteration and Loop Control Structures in C++

> **Definition to Remember**

1. Concept Overview & Fundamentals

Iteration and Loop Control Structures in C++ provides modern high-performance capabilities combining low-level hardware control with high-level abstractions, object-oriented paradigms, and Standard Template Library (STL) utilities.

In modern C++ (C++11 through C++20):

2. Technical Architecture & Memory Mechanics

C++ Runtime Execution Architecture

```text

### Modern C++ Feature Comparison

| Modern C++ Paradigm | Legacy Equivalent | Engineering Advantage | Performance Impact |

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3. Core Syntax & Implementation Patterns

Modern C++ Template Structure

```
```cpp
// Fast I/O and standard namespace configuration

std::cout << "Executing C++ Module: Iteration & Loop Control Structure" << "\n";

return 0;

```
```

4. Real-World Applications & System Engineering

- **Quantitative Finance & HFT:** High-frequency trading platforms use modern C++ for microsecond-level execution.

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5. Comprehensive Solved Problems & Code Walkthroughs

Problem 1: ArmstrongNumber

```
Problem Description & Objective:

C++ Code Implementation:

```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
#includ
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
// Output formatted data to console stream
```

```
cout <
```

```
originalNum = num;
```

```
int temp = num;
```

```
// Loop
```

```
temp = num;
```

```
// Loop
```

```
// Evaluate conditional decision logic
```

```
if (resu
```

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 2: EvenPrint

****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

### Problem 3: Factorial

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
long long calculateFactorial(int n) {
```

```
// --- Main Execution Entry Point ---
```

```
long long result = calculateFactorial(num);
```

```
// Return zero code indicating successful program finish
```

```

```

#### Problem 4: Fibonacci

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```cpp

/*

- ===

```
#include <iostream>
```

```
// Use standard library namespace globally
```

using n

```
// --- Main Execution Entry Point ---
```

int ma

```
long long t1 = 0, t2 = 1, nextTerm = 0;
```

```
// Output formatted data to console stream
```

cout <

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 5: GCD

****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
int findGCD(int a, int b) {
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

### **Problem 6: InfiniteLoop**

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Loop execution block
```

```
// Evaluate conditional decision logic
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

### **Problem 7: LCM**

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

/\*

#include <iostream>

// Use standard library namespace globally

int gcd(int a, int b) {

// --- Main Execution Entry Point ---

int lcm = (n1 / gcd(n1, n2)) \* n2;

// Output formatted data to console stream

// Return zero code indicating successful program finish

---

## Problem 8: Multiplication

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

/\*

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### **Problem 9: OddSum**

**\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
// Loop execution block
```

```
for (int
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 10: PalindromeNumber

****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
n = num;
```

```
// Loop execution block
```

```
// Evaluate conditional decision logic
```

```
// Return zero code indicating successful program finish
```

```

```

### Problem 11: PatternPrint

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Loop execution block
```

```
// Return zero code indicating successful program finish
```

```

```

### **Problem 12: PrimeNumber**

**\*\*Problem Description & Objective:\*\***

**\*\*C++ Code Implementation:\*\***

/\*

#include <iostream>

// Use standard library namespace globally

// --- Main Execution Entry Point ---

// Output formatted data to console stream

// Evaluate conditional decision logic

```
// Evaluate conditional decision logic
```

```
if (isPr
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### Problem 13: ReverseNumber

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
int original = n;
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```
---
```

Problem 14: SumOfDigit

****Problem Description & Objective:****

****C++ Code Implementation:****

```
/*
```

```
#include <iostream>

// Use standard library namespace globally

// --- Main Execution Entry Point ---

// Output formatted data to console stream

int temp = abs(n);

// Output formatted data to console stream

// Return zero code indicating successful program finish
```

```
---
```

Problem 15: SumPositive

****Problem Description & Objective:****

****C++ Code Implementation:****

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```
---
```

Problem 16: even numbers continue****Problem Description & Objective:********C++ Code Implementation:****

/*

#include <iostream>

// Use standard library namespace globally

// --- Main Execution Entry Point ---

// Return zero code indicating successful program finish

return

Problem 17: multiplication table****Problem Description & Objective:****

Implemen

****C++ Code Implementation:****

```cpp

/\*

- ===

#include &lt;iostream&gt;

// Use standard library namespace globally

using n

// --- Main Execution Entry Point ---

int mai

// Loop execution block

for (int

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### **Problem 18: positive number**

**\*\*Problem Description & Objective:\*\***

Implem

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
int mai
```

```
do {
```

```
// Outp
```

```
// Output formatted data to console stream
```

```
cout <<
```

```
// Return zero code indicating successful program finish
```

```
return
```

```
---
```

Problem 19: prime number

****Problem Description & Objective:****

Implem

****C++ Code Implementation:****

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using n
```

```
// --- Main Execution Entry Point ---
```

```
// Output formatted data to console stream
```

```
// Evaluate conditional decision logic
```

```
// Evaluate conditional decision logic
```

```
// Return zero code indicating successful program finish
```

```

```

### **Problem 20: square break loop**

```
Problem Description & Objective:
```

```
C++ Code Implementation:
```

```
/*
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
// --- Main Execution Entry Point ---
```

```
// Loop execution block
```

```
// Evaluate conditional decision logic
```

```
// Output formatted data to console stream
```

```
// Return zero code indicating successful program finish
```

```

```

**Problem 21: sum positive continue****\*\*Problem Description & Objective:\*\*****\*\*C++ Code Implementation:\*\***

/\*

#include &lt;iostream&gt;

// Use standard library namespace globally

// --- Main Execution Entry Point ---

// Output formatted data to console stream

// Loop execution block

```
// Evaluate conditional decision logic
```

```
if (num
```

```
totalSum += num;
```

```
}
```

```
// Output formatted data to console stream
```

```
cout <
```

```
// Return zero code indicating successful program finish
```

```
return
```

```

```

### Problem 22: sum until zero

**\*\*Problem Description & Objective:\*\***

Implemen

**\*\*C++ Code Implementation:\*\***

```
```cpp
```

```
/*
```

```
- ===
```

```
#include <iostream>
```

```
// Use standard library namespace globally
```

```
using namespace std;
```

```
// --- Main Execution Entry Point ---
```

```
int main()
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
do {
```

```
// Read input
```

```
// Output formatted data to console stream
```

```
cout << endl;
```

```
// Return zero code indicating successful program finish
```

```
return 0;
```

```
---
```

6. Exam Scoring Points & Memory Revision

> **Must-Write Points**

> 1. Lo

> **Quick Recall**

> `Ent